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Elementary Proofs of Ring Commutativity Theorems

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Abstract. Jacobson's commutativity theorem says that a ring is commutative if, for each x, $x^n = x$ for some $n > 1$. Herstein's generalization says that the condition can be weakened to $x^n - x$ being central. In both theorems, n may depend on x. In this paper, in certain cases where n is a fixed constant, we find equational proofs of each theorem. For the odd exponent cases $n = 2k + 1$ of Jacobson's theorem, our main tool is a lemma stating that for each x, x^k is central. For Herstein's theorem, we consider the cases $n = 4$ and $n = 8$, obtaining proofs with the assistance of the automated theorem prover Prover9.

1. Introduction

N. Jacobson's celebrated commutativity theorem for rings [14] and its generalization by I. N. Herstein [12] state:

Jacobson's Theorem. *Let R be a ring such that for each $x \in R$, there exists an integer $n = n(x) > 1$ such that $x^n = x$. Then R is commutative.*

Herstein's Theorem. *Let R be a ring such that for each $x \in R$, there exists an integer $n = n(x) > 1$ such that $[x^n - x, y] = 0$ for all $y \in R$. Then R is commutative.*

Jacobson's Theorem is a generalization of Wedderburn's "Little" Theorem that every finite division ring is commutative. Rings satisfying the hypothesis of Jacobson's Theorem have been given various names, such as *potent* rings [2, 23], J-rings [13], and probably others of which we are unaware. Choosing the first one, Jacobson's Theorem can be stated succinctly as *potent rings are commutative.* Rings satisfying the hypothesis of Herstein's Theorem seem to have never been given a separate name as far as we know, probably because the hypothesis is both necessary and sufficient for a ring to be commutative.

As noted in the statements, both theorems allow the exponent n of the power x^n to depend on x. For Jacobson's, there are various proofs in the literature; perhaps the nicest was given independently by J. W. Wamsley [25] and T. Nagahara and H. Tominaga [22].

In this paper we are interested in both theorems in the case where n is a fixed integer not depending on x, that is, rings R for which there exists an integer $n \geq 2$ such that $x^n = x$ for all $x \in R$. There have been various names suggested for such rings: J-rings [17, 18] (also used, as noted, for potent rings [13]), Jacobson rings [10], n-rings [6], n-potent rings [1], and, again, probably others of which we are unaware. Since elements x satisfying $x^n = x$ for some n are often called n-potent elements, we choose the name *n-potent* rings. We will primarily use the name in the interest of simplifying theorem statements.

The well known class of *Boolean* rings coincides with what we are calling 2-potent rings. Jacobson's Theorem for such rings is a standard exercise with an easy equational

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proof (see the first alternative proof to Theorem 2.6 below). Indeed, it is a consequence of Birkhoff's Completeness Theorem for Equational Logic [5] that for each fixed n, an equational proof of Jacobson's Theorem exists.

Among the papers devoted to proofs of Jacobson's Theorem for particular fixed n [8, 11, 26, 27], we would particularly like to single out the *tour de force* of Y. Morita [21], who gave human equational proofs for most even numbers ≤ 50 and all odd numbers ≤ 25.

There is certainly similar interest in equational proofs of Herstein's Theorem for fixed n, although the literature is not as extensive; see, e.g., [7, 19].

It is natural to try to use computer assistance to generate equational proofs of either theorem in the case of fixed n [24, 26, 27]. For Jacobson's Theorem, M. Brandenburg [6] has recently done some very exciting work along these lines.

This paper is in two parts. In §2, we discuss Jacobson's Theorem for certain cases of small n. For n odd, our main tool is a useful result we think is new (Lemma 2.4): *if R is a $(2k+1)$-potent ring, then for all $x \in R$, x^k is central.* This result was first found by Stephen Buckley (unpublished); our proof differs from his. We illustrate how helpful the lemma is for $n = 3, 5$, and 7. We also give some (we believe) new proofs for various even n. All but one of the proofs in §2 were originally human generated; the proof of Lemma 2.2 was first found by an automated theorem prover and subsequently humanized.

In §3, we turn to Herstein's Theorem for the specific cases of $n = 2$, 4, and 8. The proofs were found with the assistance of Prover9, the automated theorem prover developed by W. McCune [20]. In fact, the beginning of the authors' collaboration was an email suggestion by the second author to the first that it would be interesting to find automated proofs of Herstein's Theorem for $n = 4$ and 8, and then to see if such proofs could be suitably humanized.

We would judge the humanization effort to be quite successful for $n = 4$ (Theorem 3.7) and somewhat successful for $n = 8$ (Theorem 3.8). For the latter proof, although it is certainly possible for a patient human to follow the individual steps, the overall pattern is difficult to see. We have no idea how or even if a general idea can be extracted from the proof which could be applied to higher powers of 2.

We conclude this introduction by discussing conventions. Rings are assumed to be associative but not assumed to have a unity, that is, they are what some, following a suggestion of Jacobson ([15], pp. 155–156), would call a "rng".

The *centre* of a ring R is the subring $Z(R) = \{a \in R : ar = ra, \forall r \in R\}$. Elements of $Z(R)$ are said to be *central*.

In our proofs, especially in §3, we will make heavy use of the commutator $[x, y] = xy - yx$. This is an interesting binary operation in its own right, but here we mainly use it as a computational tool. It turns out that introducing commutators into Prover9 input files and feeding the program basic facts about commutators helps quite a bit in finding proofs. We will discuss the commutator identities we need at the beginning of §3.

2. Jacobson's Theorem for n-potent rings

Our main interest in this section is giving equational proofs that n-potent rings are commutative for various values of n. However, for some preliminary results, there is essentially no extra work involved in giving proofs for reduced rings. A ring R is said to be *reduced* if it has no nilpotent elements. This can be equivalently described by the condition

$$x^2 = 0 \implies x = 0\,, \tag{1}$$